

Conversion of a Coaxial Rotorcraft to a UAV

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A coaxial helicopter with a maximum take-off weight of 600kg was converted to an unmanned drone as part of a research and development project at the Technical University of Munich. A minimally invasive robotic actuator system has been developed which can be retrofitted on the copilot seat of the rotorcraft in a short period of time to enable automatic flight. The automatic flight control robot includes electromechanical actuators which are connected to the cockpit inceptors and control the helicopter. Most of the sensors and avionic components are integrated into the modular robotic system for faster integration into the rotorcraft. The mechanical design of the control system, the development of the robot control software, and the control system architecture will be described in this paper. Furthermore, references to other publications describing other aspects of our work such as flight dynamics, advanced aerodynamics and flight control are provided.

I. Nomenclature

r	=	Non-dimensional rotor radius
θ_{col}	=	Helicopter collective pitch input at the $r = 0.7$ reference radial position of the rotor
θ_c, θ_s	=	Helicopter cyclic pitch inputs at the $r = 0.7$ reference radial position of the rotor
θ_{yaw}	=	Helicopter yaw input, i.e. additional collective pitch input of the upper rotor at $r = 0.7$
p, q, r	=	The roll, pitch, and yaw rates of the helicopter
ϕ, θ, ψ	=	The roll, pitch, and yaw attitude angles of the helicopter

II. Introduction

THE CoAX 600 rotorcraft, developed by edm aerotec GmbH [1] has been converted to a remotely piloted drone via a robotic actuator system, mounted onto the co-pilot seat in the cockpit. CoAX 600 is a coaxial helicopter with a Maximum Take-Off Weight (MTOW) of 600kg. It has been designed according to the German ultralight (UL) rotorcraft standards and has a UL type certificate from the German German Ultralight Association (DULV). It is a two seater rotorcraft, fitted with mechanical controls.

The advantages and use-cases of unmanned aerial vehicles of different weight classes and configurations in the civil and military contexts have become more evident in the recent years. Besides the low-cost mostly multi-rotor drones widely available to the public for photo- and videography, aerospace manufacturers have developed multiple Unmanned

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Aerial Vehicles (UAVs) with vertical take-off and landing capabilities for different applications. Kaman Corporation was recently awarded for the development of a cargo UAV quadcopter [2]. The KARGO UAV is capable of transporting a payload of up to 800 lbs over 500 nm. Other manufacturers such as Israel Aerospace Industry [3] have developed rotorcraft UAV with the main and tail rotor configuration, capable of lifting up to 180kg payload with a top speed of 150km h^{-1} . The K-Max helicopter with intermeshing rotors can be mentioned as a pioneer of unmanned helicopters designed for (external) cargo transport [4, 5]. The double rotor, in this case intermeshing rotor design is known to be more efficient than conventional main and tail rotor configuration when the flight operations mostly consist of hovering and slow flight. In a more recent project, a Black Hawk helicopter was transformed into an Optionally Piloted Vehicle (OPV) by Sikorsky Aircraft [6].

In all of the projects mentioned above, the actuators are deeply integrated in the rotorcraft mechanical control system. The rotorcraft are either designed as unmanned systems from the beginning or the manned-unmanned conversion is a long process and not easily reversible. The idea of retrofitting a robotic flight control system with integrated avionics has been investigated before. In the ROBOPilot [7] such a system was designed for and tested on a Cessna 172 aircraft. The system successfully demonstrated automatic flight. In another study a robotic actuator system was mounted into a SVH-4 rotorcraft and was tested in a hovering platform [8]. No free flight of the helicopter has been reported in this project. The control mechanism in the cockpit consisted of a multi-arm robot and additional mechanisms for controlling the pedals and the collective lever. This concept is similar to a Aurora Flight Sciences study [9] in which a multi-arm multi-purpose robot was retrofitted into a Cessna Caravan co-pilot seat and could successfully fly the aircraft.

In this paper, an overview of the development process for the robotic actuator system will be provided. The paper follows with the details of the mechanical design of the control rods, the choice of the actuators and the robot control software which cancels the nonlinear kinematics and couplings of the control system. Furthermore, the architecture of the flight control system (FCS), enabling unmanned operations, will be described in this paper.

III. The Control Mechanism

The robotic system consists of three separate mechanisms, controlling the stick, collective lever and the pedals. The stick motion is controlled by two servos, sitting on top of the shortened stick in the orientation depicted in Fig. 1. Simultaneous actuation by both servos is required for a linear motion of the stick in any direction, commanded by the flight control laws. This combined actuation is realized by the robot control software which cancels the nonlinear kinematics of the mechanism. The robot control software generates commands for each actuator, to realize the commanded flight controls. The actuator commands are computed using the inverse kinematics of the robotic system, which is integrated in the robot controller software as multi-dimensional look-up tables. The robot control software is described in Sec. III.B in detail.

The development of the mechanical control hardware and software was supported by a multi-body, multi-domain simulation model of the robotic flight control system, developed in Simulink, based on the developed CAD models of the robot integrated in the rotorcraft. The look-up tables, determined from the multi-body simulation model were corrected with limited measurements on the integrated robot in the helicopter copilot seat.

The pedal and collective lever are also each controlled by a servo as depicted in Figs. 2a and 2b. The nonlinear control kinematics in this case is also canceled with inverse kinematics look-up tables embedded into the robot control software. However, the look-up tables are one dimensional for the collective lever and the pedal and can be easily determined based on a multi-body simulation model and corrected using measurements on the real system.

All of the four servos on the rotorcraft are of the type DA72, developed by VOLZ Servos GmbH. The mechanical levers are manufactured from aerospace-grade aluminum and stainless steel. High precision machining has been utilized for manufacturing such that no backlash could be noticed in the robot joints. The dual-motor servos also include an anti-backlash function which uses force-fighting between two motors to significantly reduce any backlash in the actuators to 0.1 deg or less.

A. Design of the Robotic Control System and the Multi-body Simulation

The robotic flight control system was developed in the CAD software Catia. The design needed to satisfy the following requirements:

- Covering the complete motion space of the cockpit inceptors,
- Avoiding singularities in the motion space,
- Meeting the actuation rate requirements,
- Satisfying the required forces needed for the control of the robot, and

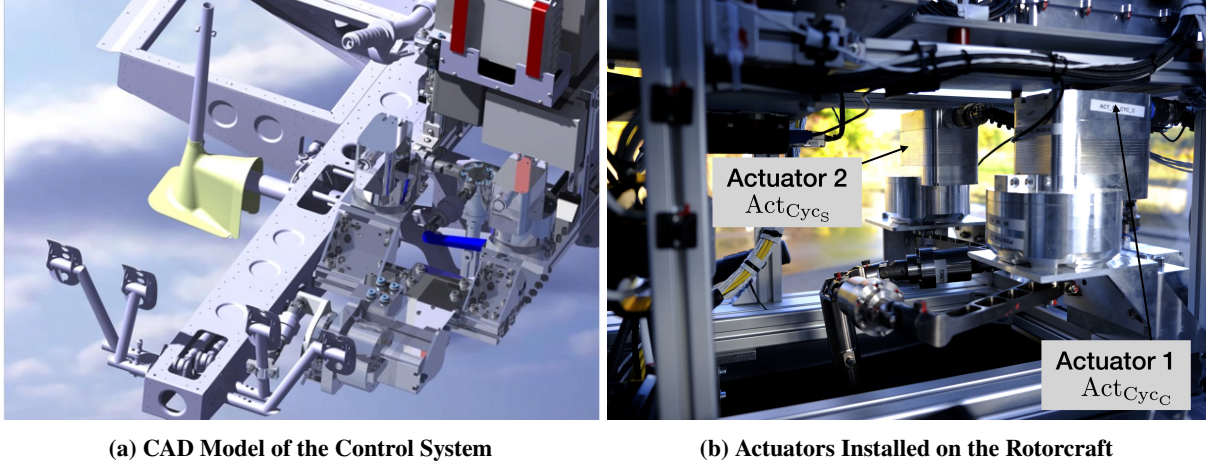


Fig. 1 The actuators controlling the stick.

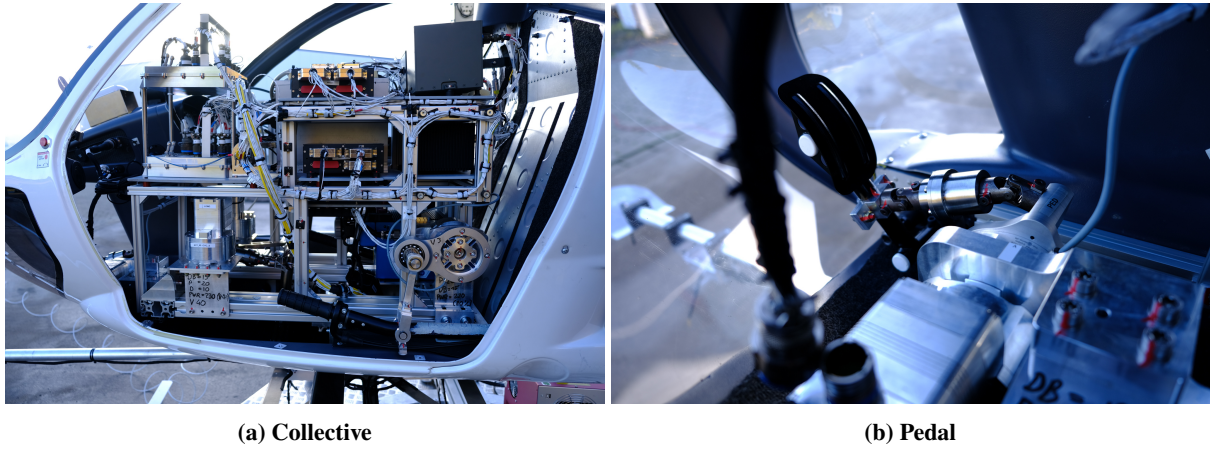


Fig. 2 Actuators controlling the collective lever and the pedal.

- The safe operation of the custom-designed parts, as well as the chosen norm parts.

A multi-body simulation model was developed based on the CAD-models already in the early stages of the design and development process, to ensure that the designed kinematics overall and each individual part fulfill the requirements mentioned above. Furthermore, by augmenting the multi-body simulation model of the control kinematics with physics-based models for the actuators, the development of the actuators could be supported, since they were being developed by VOLZ Serovs GmbH during the project.

The toolbox ProSys [10], developed at the institute of Flight System Dynamics of TU Munich was utilized to set up the multi-body simulation model. Prosys can automatically import the system dynamic and kinematic information of a mechanism from CATIA into Matlab and set up a Simulink-Simscape model, if certain guidelines are followed in the development of the CATIA model, such as the mechanism being modeled as a DMU kinematics module. The imported data include the information about the system mechanical joints (type, position, degree of freedom), as well as the mass, center of gravity position and the moments of inertia of each part. The automatically generated multi-body simulation model does not have any actuation and is by default simulated only under the influence Earth's gravity.

Simscape allows for each joint, available in multi-body model to be actuated. This has been used to actuate the mechanism using the joints which represent the connection of an arm to a servo. If the rotational motion is commanded at a rotational joint, the required torque to realize the commanded motion can be measured at the same joint. This interface was also utilized to couple the simscape multi-body model with physics-based models of the actuators in Simulink. Fig. 3 shows the coupling of the servo simulink model with the revolute joint, modeling the connection of the collective servo arm with the servo shaft, as an example. The actuator models were parametrized based on the initial

configuration of the servos (choice of motor, motor controller, electronics, and gearbox). This creates the simulation model for the forward kinematics of the robot.

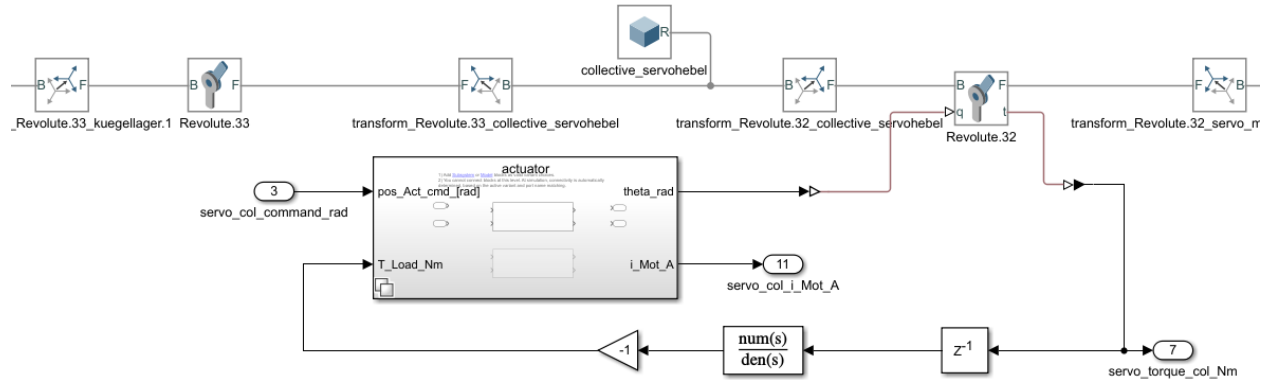


Fig. 3 Coupling of a servo model with a revolute joint.

The same base non-actuated model has been used to develop a model for the inverse kinematics of the robot. In the inverse kinematics subsystem, the joints representing the position of the cockpit inceptors (cyclic stick, collective lever, and the pedal) were used for actuation of the multi-body simulation model. The revolute joint at the respective servo shafts rotate according to the commanded cockpit inceptor position such that the boundary conditions of the kinematics are fulfilled. The motion of the revolute joints at the servo shafts is tapped and used as input for the servo models of the forward kinematics. Fig. 4 provides an overview of the overall simulation set up. It shows how the same base auto-generated multi-body model is utilized to create the inverse and forward dynamics/kinematics models. The commands into the forward dynamics model, which also includes actuator dynamics are low-pass filtered (with a cutoff frequency of 100Hz), since abrupt changes in the commands drive the linear algebraic system into a singularity.

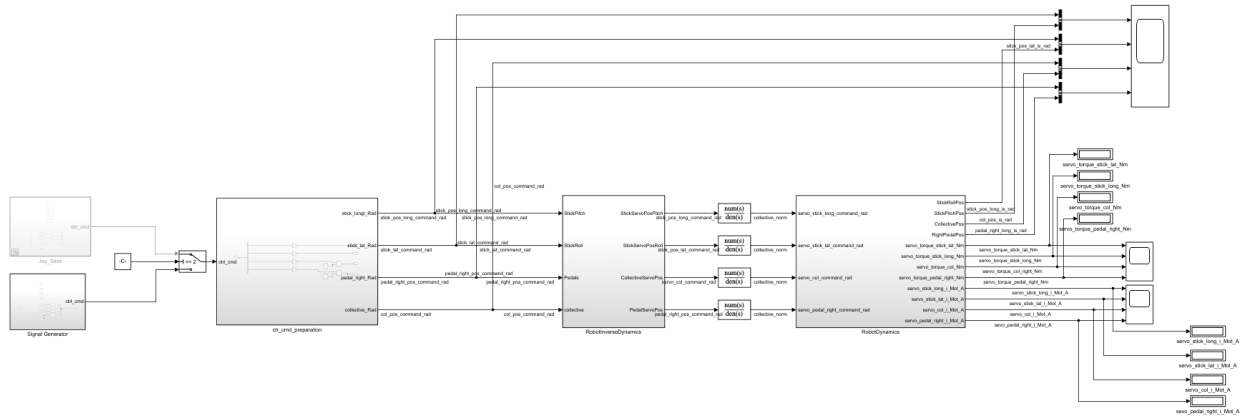


Fig. 4 Structure of the simulation framework for the robot controller.

The revolute joints, connecting the cockpit inceptors and the helicopter rigid body were augmented with internal dynamics in form of a spring-damper system to represent the aerodynamic loads occurring during the flights. The spring and damper coefficients have been chosen according to aerodynamic load measurements during manned flight tests. The actual position of the cockpit inceptor joints, achieved with the actuation of the servos under load, as well as the moments acting on the servos, and the servo power usage are provided as the outputs of the overall simulation model.

B. Robot Control Software

Each servo has its own position controller, designed by the manufacturer, VOLZ servos GmbH. However, a feed-forward inverse-kinematics-controller is needed to generate position commands for the servos which would result in

a specific position of a cockpit inceptor, matching a blade pitch angle command, generated by the flight control law. The simulation setup, presented in sec. III.A already includes the inverse kinematics of the robot. The complex Simscape model is however not suitable for deployment on a flight control computer. The simulation of such multi-body models in real-time requires computational resources which exceed those of a typical flight control computer. Furthermore, solving such linear algebraic equations is not deterministic and the execution time is difficult to predict. Therefore, the inverse kinematics of the robot was integrated in the robot control software as one dimensional Look-up-Tables (LuTs) for the pedal and the collective lever, and two dimensional LuTs for the control stick.

The look up tables were determined by simulating the inverse kinematics multi-body simulation for a 45x45 uniform grid of the cyclic stick in equidistant longitudinal and lateral positions. The corresponding positions of the servos $\text{Act}_{\text{Cyc}_C}$ (actuator 1) and $\text{Act}_{\text{Cyc}_S}$ (actuator 2) are recorded during 2025 simulations of the multi-body simulation model at the 45x45 cyclic stick position. Limited adjustment of the LuTs based on measurements on the real system were necessary due to the manufacturing tolerances and further inaccuracies in the assembly of the robot in the helicopter. The servo internal position measurements using a hall sensor and the angles of the control stick, measured by inclinometers were used for the calibration of the LuTs. The longitudinal and lateral calibration points, as well as the full inverse kinematics LuTs are visualized in Figs. 5 and 6.

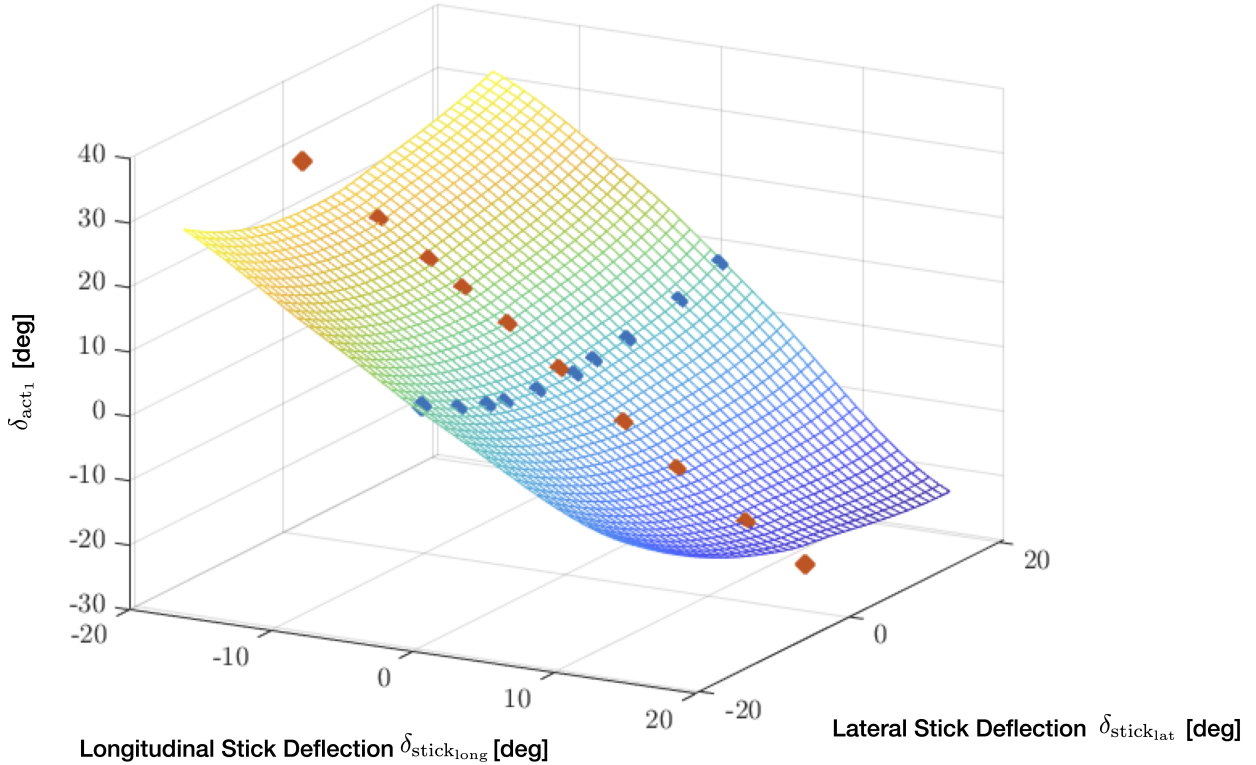


Fig. 5 $\text{Act}_{\text{Cyc}_C}$ (Actuator 1) inverse kinematics grid.

A linear interpolation scheme has been applied to the points of the 45x45 grid σ_{sim_i} for each actuator ($i \in \{1, 2\}$), denoted by $v(\delta_{\text{stick}_{\text{long}}}, \delta_{\text{stick}_{\text{lat}}}, \sigma_{\text{sim}_i})$, which can be evaluated at any stick position query point $[\delta_{\text{stick}_{\text{long}}}, \delta_{\text{stick}_{\text{lat}}}]$ to determine the output shaft position of each of the actuators (δ_{act_i}) required to achieve a desired stick position. The grid for each actuator consists of the vectors of the longitudinal and lateral stick positions at which the virtual measurements took place (α_{long} and α_{lat}), and the corresponding position of the servo output shaft $\delta_{\text{act}_{i\alpha}}$

$$\sigma_{\text{sim}_i} := \alpha_{\text{long}} \times \alpha_{\text{lat}} \rightarrow \delta_{\text{act}_{i\alpha}}, \quad i \in \{1, 2\} \quad (1)$$

The initial simulation-based uncorrected grid was evaluated at the measured stick positions to compute the error between the simulation-based grid and the longitudinal measurement points (blue in Figs. 5 and 6), denoted as λ

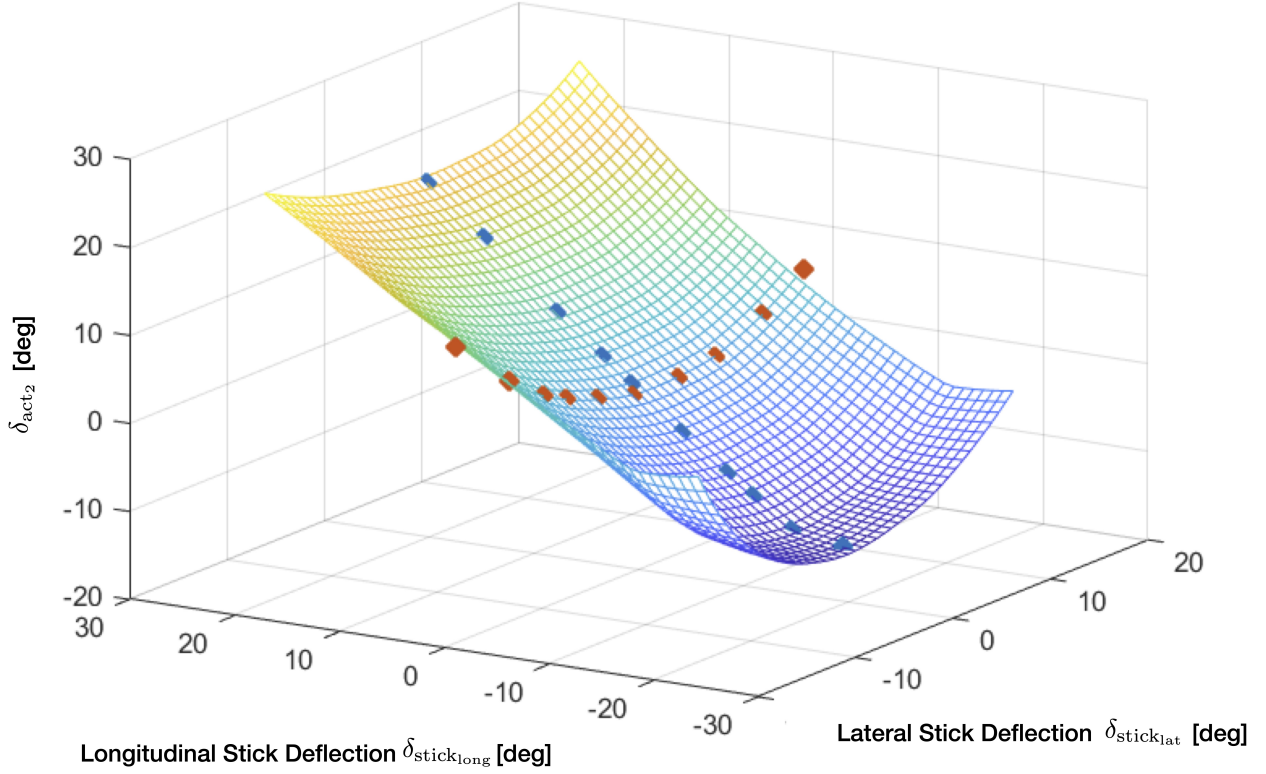


Fig. 6 Act_{Cycs} (Actuator 2) inverse kinematics grid.

$$e_{i\lambda} = \delta_{act_{i\lambda}} - v(\lambda_{long}, \lambda_{lat}, \sigma_{sim}), \quad i \in \{1, 2\} \quad (2)$$

where $\delta_{act_{i\lambda}}$ is the measured position of the stick actuators 1 and 2. $\lambda_{long}, \lambda_{lat}$ are the longitudinal and lateral deflections of the stick at which the actuator position was sampled. The chosen longitudinal positions of the stick were commanded via the initial simulation-based inverse kinematics grid. Therefore, any measured nonzero λ_{lat} is a result of the error in the inverse kinematics grid. We can form a longitudinal error grid for each actuator with $e_{i\lambda}$, and the sampling points λ

$$\sigma_{e_{i\lambda}} := \lambda_{long} \times \lambda_{lat} \rightarrow e_{i\lambda}, \quad i \in \{1, 2\} \quad (3)$$

In a similar way, the grid error for the lateral measurement points β (red points in Figs. 5 and 6) is computed after the application of the longitudinal correction, using the above error grid $\sigma_{e_{i\lambda}}$ of the longitudinal measurement points and the two-dimensional linear interpolation operator v

$$e_{i\beta} = \delta_{act_{i\beta}} - v(\beta_{long}, \beta_{lat}, \sigma_{sim}) - v(\beta_{long}, 0, \sigma_{e_{i\lambda}}), \quad i \in \{1, 2\} \quad (4)$$

The lateral measurement points can be collected together with their associated error to form the grid

$$\sigma_{e_{i\beta}} := \beta_{long} \times \beta_{lat} \rightarrow e_{i\beta} \quad (5)$$

for the correction of the inverse kinematics grids with respect to the stick lateral deflection measurement points. The feed-forward control law becomes

$$\delta_{act_i} = v_{sim}(\delta_{stick_{long}}, \delta_{stick_{lat}}, \sigma_{sim}) + v(\delta_{stick_{long}}, 0, \sigma_{e_{i\lambda}}) + v(0, \delta_{stick_{long}}, \sigma_{e_{i\beta}}) \quad (6)$$

The one dimensional LuTs for the pedal and collective lever are developed only based on the direct measurements on the system, as enough points in the motion space could be measured in the one dimensional case. Figs. 7 and 8 show

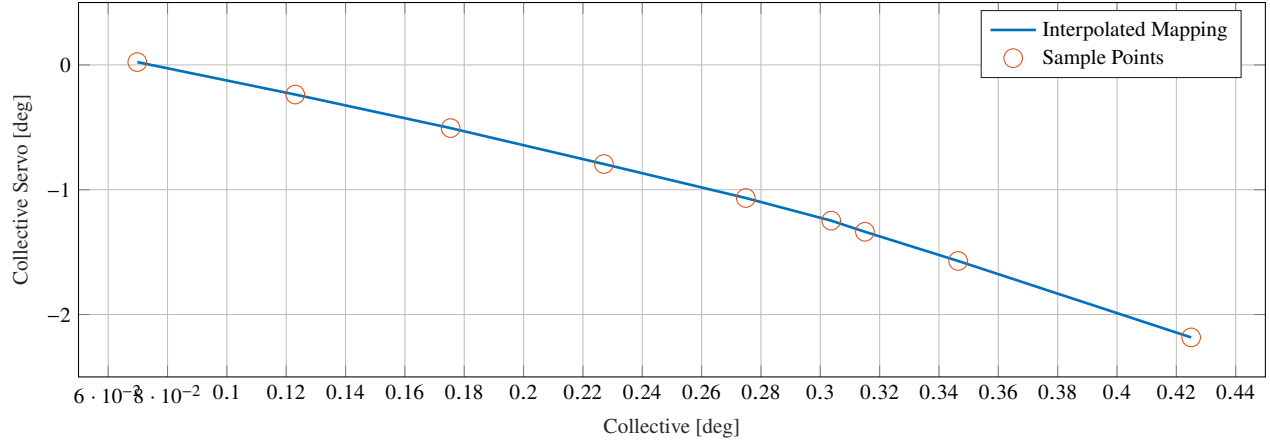


Fig. 7 Collective lever actuator inverse kinematics grid.

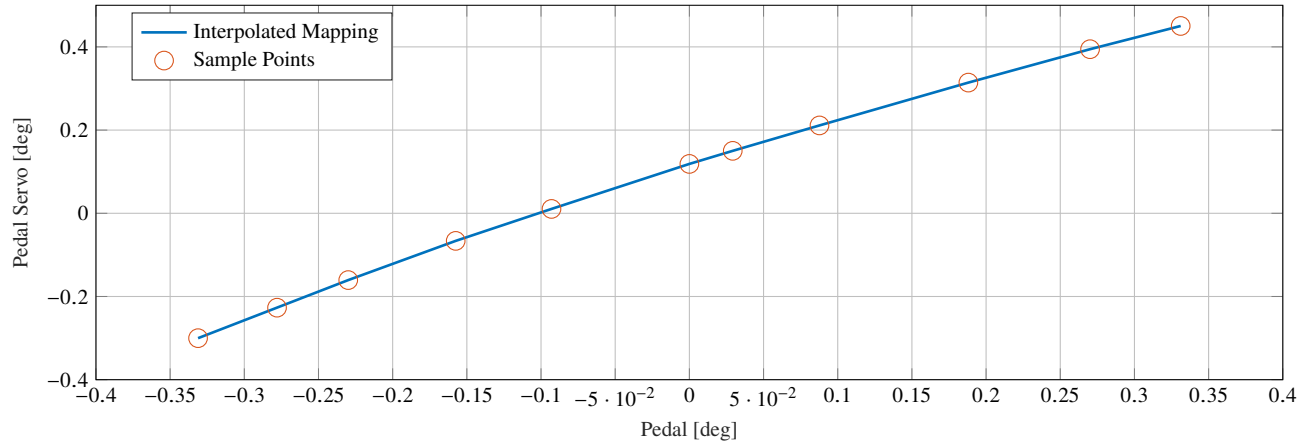


Fig. 8 Pedal actuator inverse kinematics grid.

the mapping between the collective and pedal positions and their respective actuators. Nonlinearities are evident in all of the one and two dimensional mappings.

The structure of the robot control software is provided in Fig. 9. The collective and cyclic commands, computed by the flight control law are converted to cockpit inceptor positions by a first set of LuTs. The LuTs of Figs. 5, 6, 7, and 8 are applied to the cockpit inceptor positions to compute the corresponding servo positions.

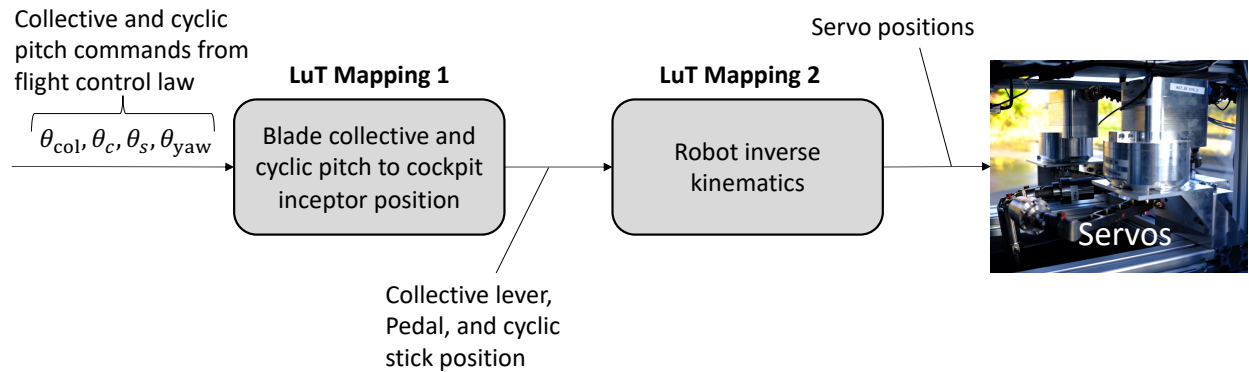


Fig. 9 APCU Software structure.

C. Integration and Tests

In a first step, the flight control robot was assembled on a stationary frame of the helicopter in the lab. The frame had all of the original dimensions and the included mechanical controls of the rotorcraft (cf. Fig. 10). The concept, and the different functions of the robot were tested and validated in the lab. The setup was also used to tune the actuators internal position controller.

The setup has been extended by the hardware-in-the-loop (HIL) simulation of the helicopter. The simulation model of [11] was used for this purpose. The HIL-tests included the simulation model, the flight control computer, the actuators, and the pilot, controlling the helicopter via a hand-held radio control (RC) transmitter. The hardware setup can be seen in Fig. 10 and the interaction of the different hardware and software components of the tests is visualized in Fig. 11. The control commands from the RC are received by multiple receiver units, which forward the received command data to two control units (PDCU and FCC, for details refer to section III). Depending on the active control mode, the unit in command computes rotor blade angle commands, which are transmitted to a separate actuator control unit (APCU) running the robot control software. The servo position is measured by the integrated hall sensors and transmitted to the realtime PC running the simulation model. The simulation model consists of the flight dynamics of [11], and models for the forward kinematics of the control system including the robot (mapping from servo positions to rotor collective/cyclic pitch), as well as sensor models, among others for the attitude heading reference system.

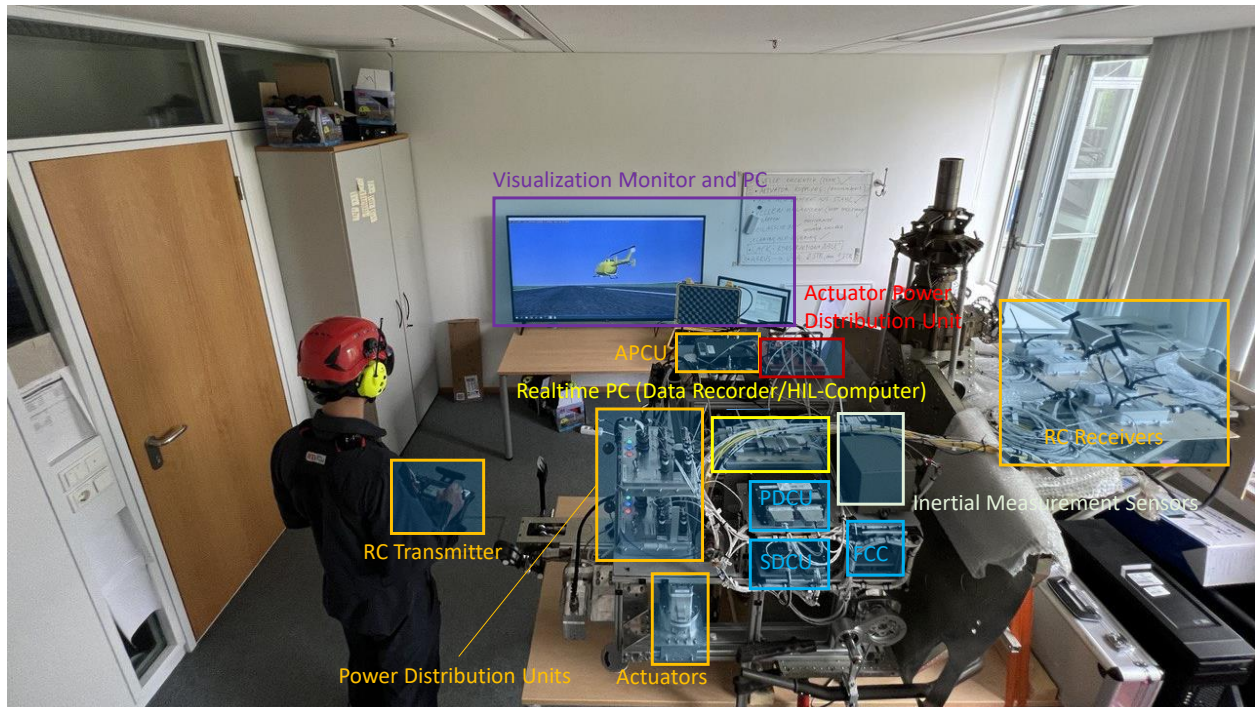


Fig. 10 System setup for HIL tests.

In a second step, the robotic controller, including the electronic flight control system were integrated into an electric variant of the CoAX 600 rotorcraft which was mounted in a hover frame, allowing for the rotation of the helicopter around the mounting point. The whole hover frame can also move on wheels which allows for the translational motion of the rotorcraft, however with significant wheel friction. The flight control robot and the electronic control system was exposed to the real-world helicopter flight conditions, with a rotating rotor in the hover frame for the first time. The cockpit inceptors were equipped with Mechanical locks to fix the controls during the engine start up process. This countermeasure was needed as to prevent the actuators to become unstable, or drift due to the inertial and aerodynamic loads (of a wide frequency spectrum) during the rotor start up. The cockpit inceptor locks were released starting with the pedal, collective and finally the cyclic stick. The system was operated in a direct control law without artificial feedback (hereafter called direct law) during the first tests and the pilot controls were at the neutral position (zero cyclic, and yaw inputs, with collective lever in the lowermost position). The first flight in the hover frame was conducted subsequent to the safe operation of the robot in the hover frame with the rotating rotor on the ground (Fig. 18). Testing the concept,

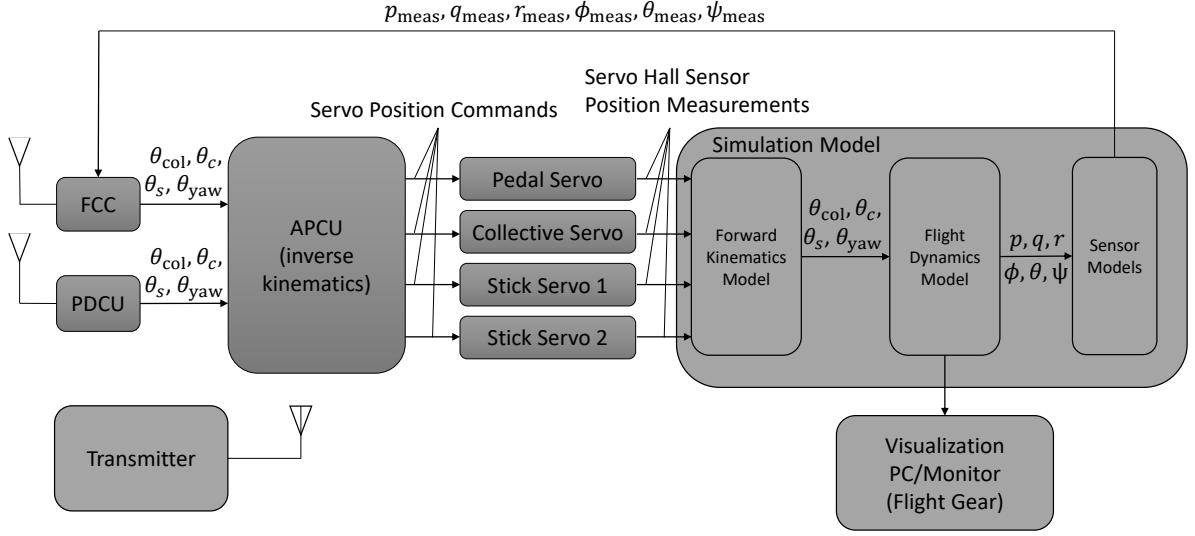


Fig. 11 HIL tests block diagram

the mechanical system, the avionics and the team on the hover frame was of utmost value as preparation for free flights.

The opportunity of flight tests with the helicopter on the hover frame was used to gather data for the identification of low order equivalent systems for the robotic actuator system. Such simple linear models are essential for the stability and robustness analysis of the flight control law. The data was generated using automatic injection of doublet, sweep, and multisine maneuvers. This process will be described in Sec. III.D.

The flight test campaign on the hover frame stretched over 4 months and included testing the initial versions of the different modes of the flight control software, described in [11]. The flight control robot and the avionics were integrated in a regular CoAX 600 after the completion of the flight test campaign on the hover frame. The integration of the modular flight control system into a new helicopter takes ca. 2 weeks of work, based on our experience.

The calibration of the two-dimensional inverse kinematics grids for the two stick actuators (Figs. 5 and 6) and the measurement of the one-dimensional curves for the collective and pedal servos (Figs. 7 and 8) should be repeated after the integration of the robot into a new helicopter, as even slight changes in the position of the rack, carrying the actuators would result in the misalignment of the stick motion, if the grids are not corrected. The misalignment error means that a pure longitudinal or lateral stick command results in a diagonal stick motion.

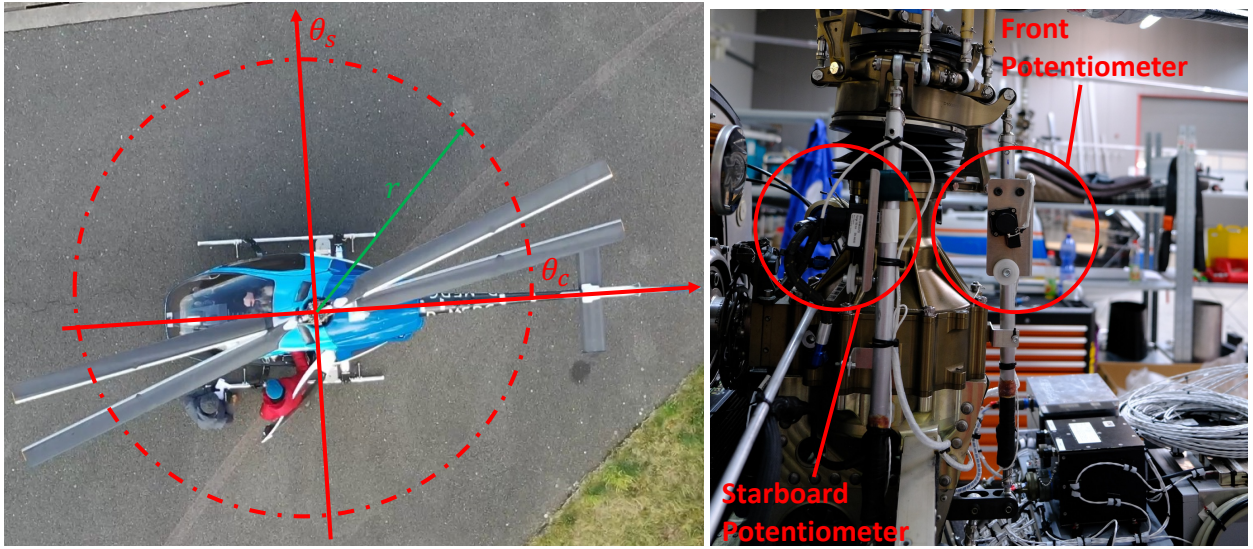
D. Low Order Equivalent Systems for the Actuator Dynamics

Linear Low Order Equivalent System (LOES) models were needed for the stability and robustness analysis of the flight controller, as described in [11]. These linear differential equations provide a simplified representation of the dynamics between a commanded rotor collective or cyclic pitch by the flight control law ($\theta_{colcmd}, \theta_{ccmd}, \theta_{scmd}, \theta_{yawcmd}$) to actual blade pitch angle, achieved by the the actuators. The blades have a linear twist. Therefore, the pitch angle at the radial position of $r = 0.7$, with r being the non-dimensional rotor radius, is chosen as the reference point for the blade pitch commands and measurements.

This could be done by linearization of the nonlinear model of the robotic system in Simulink/Simscape (cf. Sec. III.A) and applying model order reduction methods to achieve a linear low order model representing the onboard actuation system. A fully simulation-based approach can however, not account for all of the real-world effects, including the mentioned misalignment errors. Therefore, specific tests were performed on the electric helicopter in a hover frame (cf. Sec. V) to gather data for identifying the LOES from flight data. A series of sine sweep and multisine maneuvers as described in [12] were automatically injected during hover flights in the hover frame. The maneuvers were in the

frequency range, in which the LOES poles were expected to be. The maximum sweep/multisine frequencies were limited by the 50Hz sampling rate of the data acquisition system.

The true in-flight actuator responses were measured indirectly using four draw-wire potentiometers. The lower rotor pitch angles were chosen as the reference response to the cyclic and collective commands. The yaw command only affects the upper rotor. Three of the potentiometers were mounted onto the three control rods on the starboard and port sides, as well as the front of the swashplate. The fourth potentiometer was positioned on one of the control rods connecting the pedals to the upper rotor swashplate and the rudder. The rotor pitch angle at the $r = 0.7$ position was measured using inclinometers, mounted on each blade on the ground. The potentiometer readings were also recorded for each of these measurement points. The test was repeated for multiple azimuth angles and multiple settings of the cockpit inceptors to form a measurement grid. A linear function, relating the potentiometer readings to the lower rotor collective and cyclic pitch angles θ_{col} , θ_c , θ_s , as well as the upper rotor additional collective for yaw control θ_{yaw} was identified with the help of least squares methods. This linear function was applied to the potentiometer measurements captured during the flight tests in the hover frame to reconstruct the rotor collective and cyclic pitch responses. Two of the wire-draw potentiometers and the measurement position of the θ_c and θ_s measurements on the lower rotor are visualized in Fig. 12.



(a) Reference position of the rotor blade pitch angle measurements/commands. (b) Draw-wire potentiometers for blade pitch position measurements.

Fig. 12 Rotor cyclic controls reference positions and swashplate instrumentation.

The estimation of the actuator models took place in the frequency domain using the Matlab system identification toolbox. Three transfer functions $\Theta = G_\theta \Theta_{cmd}$ were estimated for the input channels θ_s , θ_c , and θ_{yaw} . It was observed that the data from the potentiometer on the swashplate front position was extremely noisy. This noise mostly affected the pseudo-measurements of θ_s . Due to this extreme noise, the system identification results for the G_{θ_s} transfer function (for the longitudinal stick deflection) were poor. Therefore, only an actuator dynamic for the lateral stick motion was estimated. Due to the mostly similar kinematics and based on the analysis of ground data, the assumption was made that the actuator dynamics in the stick longitudinal and lateral motion are similar ($G_{\theta_s} \approx G_{\theta_c}$). No actuator model for the collective lever has been estimated since this channel is not controlled with a feedback controller at this stage of the work.

First and second order linear model structures were tested for parameter estimation. Furthermore, a signal transport delay of 0.02s was witnessed in the data and considered in the model structure. The following first-order transfer function was estimated for the stick cyclic commands

$$\Theta_{c/s} = e^{-0.02s} \frac{19.37}{s + 21.79} \Theta_{c/scmd} \quad (7)$$

and the yaw actuator dynamics is characterized by the following second-order system

$$\Theta_{yaw} = e^{-0.02s} \frac{661.4}{s^2 + 37.37s + 654.4} \Theta_{yaw_{cmd}} \quad (8)$$

The stick actuator has a single pole at 21.8 rad s^{-1} and the pedal actuator has a stable pole pair at 25.6 rad s^{-1} , with a relative damping of 0.73. The poles can be seen in Fig. 13. The model fit in the frequency domain can be seen in Fig. 14.

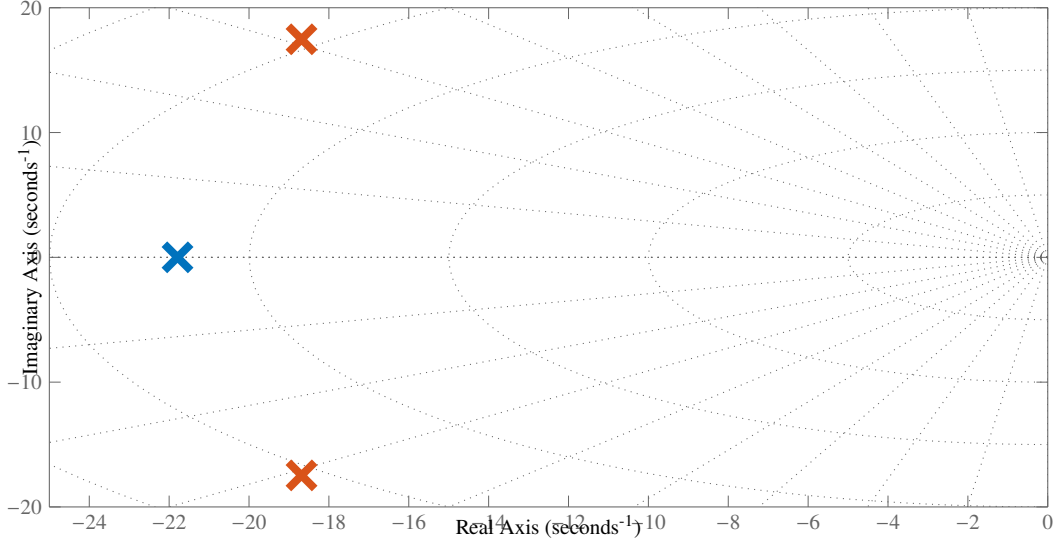


Fig. 13 Actuator dynamics poles.

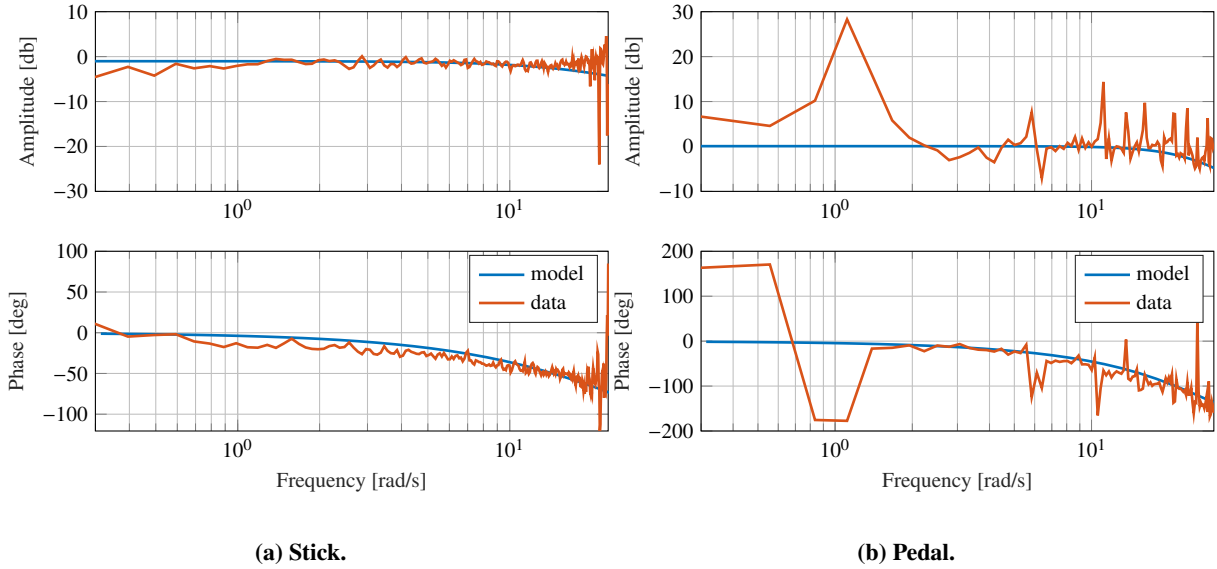


Fig. 14 Actuator model fit in the frequency domain.

The identification of the actuator models was made difficult by the low sampling rate of the data, and the amplitude of the excitations. Automatic maneuver injection at higher frequencies and with higher amplitudes were designated as too dangerous to conduct on the helicopter in the hover frame, and with the rotating rotor. Fig. 14 shows an overall good agreement with the data. These models were used for stability and robustness analysis of the controllers. Therefore, safety factors have been applied to the actuator bandwidth in [11] to compute the stability margins, to account for any possible errors in the determination of the actuator models.

IV. Flight Control System

A. Operational Concept and Requirements

The concept for the flight control system (FCS) is driven by the intended application as an unmanned technology demonstrator for the robotic actuation system. The main required functions of the system are:

- **Manual remote control:** During the system definition it was decided to implement only manual control in a first step in order to reduce the complexity for the demonstration flights. The manual remote control system also serves as a basis for a potential later extension with automatic flight control functions. In this case, fallback to manual control would provide an emergency procedure to mitigate malfunctions of the automatic flight control system.
- **Telemetry data indication:** The system must provide the capability for the remote crew to monitor critical system parameters in realtime on the ground during operations.
- **Remote flight termination:** The possibility to remotely shutdown the engine is required as a safety measure to ensure containment of the UAV within a predefined geographical volume, thereby reducing the risk to other airspace participants or third parties on the ground in the event of a control loss.
- **Measurement data collection for post-flight processing:** A wide array of sensors has been mounted on the rotorcraft, including two inertial measurement units at different positions, two GPS receivers, an angular acceleration sensor, and potentiometers and strain gauges at different positions along the control rods.

Additional safety objectives result from the application of the SORA (Specific Operations Risk Assessment, see [13, Article 11]) process, which is required for the operational authorization of the rotorcraft as an unmanned aircraft system (UAS) by the German authorities. Following this process, objectives are defined based on a specific assurance and integrity level (SAIL). The SAIL is a qualitative measure that drives the objectives to be fulfilled by an applicant. It is defined based on an assessment of the risk of the UAS operation on the ground and in the air (for details refer to [13, AMC 1 to Article 11]). As we choose to operate the rotorcraft over a so-called "controlled ground area" where access is restricted to members of the remote crew, operate only within visual line-of-sight (VLOS) and reduce the risk of encountering another aircraft by operating in an active aerodrome traffic zone (ATZ), which is exclusively used by the UAS, the operation is classified in SAIL II. In this context, two fundamental safety requirements are given by [13]:

- **Containment:** *"No probable failure of the UAS or any external system supporting the operation should lead to operation outside the operational volume."* [13, p. 61]
- **Safety and Reliability:** *"The resulting hazards are minimized in the event of a probable malfunction or failure of the UAS."* [13, p. 109]

The first objective requires to contain the operations to a predefined, approved geographic volume and addresses the risk resulting from exceedance of this operating volume as a result of loss of control over the UAV. Note that compliance to the second requirement is optional for SAIL II operations but it was decided to use this as a governing principle for the system architecture design as far as practicable. As a basic design principle, fault tolerance against any single failure, unless it can be reasonably assumed to be improbable, was required for the presented system architecture as a result of this objective.

B. System Architecture

A schematic overview of the flight control system is shown in Fig. 15. The aircraft is always - in all operating modes - controlled remotely by the remote pilot, using a hand-held remote control (RC). Furthermore, a second remote pilot has the possibility to take over control using a backup RC. A ground monitoring unit (GMU) provides live indication of telemetry data. Onboard the air vehicle, two separate flight control channels process the commands received from the remote pilot: A primary flight control system (PFCS) provides active closed-loop control of the rotorcraft and additionally contains the equipment required for measurement data acquisition and the telemetry data link. The direct flight control system (DFCS) acts as a backup system and provides a direct control mode in which the commanded blade pitch angles of the lower and upper rotor and the rudder deflection are directly proportional to the control stick deflections on the remote control, without any artificial feedback loops. Each flight control channel has its own, independent power supply as well as its own set of receiver units to process RC signals.

A separate dual-channel control unit (APCU) executes the robot control software and translates the flight control commands into direct servo deflection orders. The electro-mechanical servos feature a duplex design, where the output shaft is driven by two electric motors that are combined using a torque summing gearbox. Each electric motor is driven by its own motor control unit, that is connected to one of the two APCU channels.

The detailed electronic architecture is shown in Fig. 16. The remote control functions are executed on two

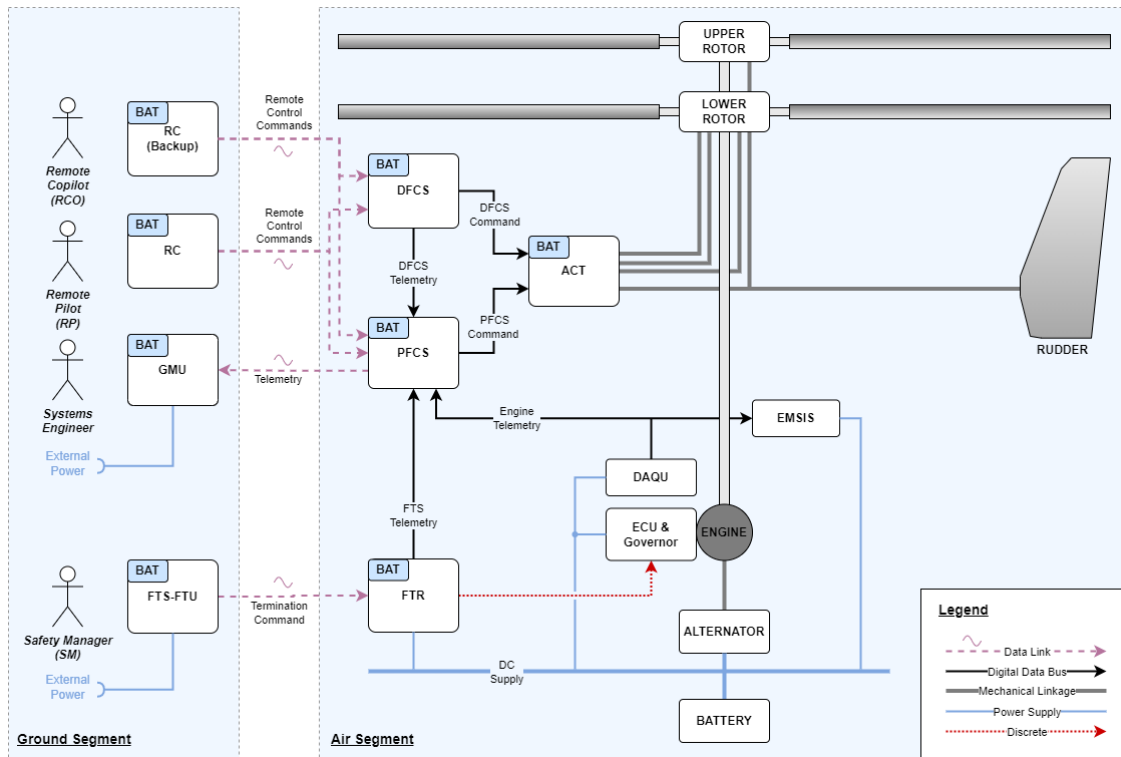


Fig. 15 Flight control system schematic overview

control units: The flight control computer (FCC) provides closed loop attitude and angular rate control. The inertial measurements for these control functions are captured by an Air Data Attitude Heading Reference (ADAHRS) unit. Note that no air data measurements are used for closed loop control. The backup direct mode control is executed by the so-called "Pilot-DCU" (PDDU). Each control unit receives command signals from the RC through multiple receiver units (called PRX in Fig. 16). The receiver units are placed on different locations outside the helicopter fuselage to ensure stable reception on multiple units in any orientation of the helicopter. A cross-talk interface between PDCU and FCC additionally forwards the control commands received by the PDCU to the FCC such that the augmented control through the FCC remains available even if reception is lost on all of the directly connected receiver units.

The fallback to the DFCS is implemented by a simple priority selection logic on the actuator control unit: In normal operating conditions, the PDCU is passive and the FCC is in command. Whenever the PDCU is transmitting commands, the PDCU commands are selected by the APCU channels. The PDCU can be activated either manually (using dedicated switches on the RC) or automatically, when a loss of commands from the FCC is detected. For this purpose the actuator control unit monitors the periodic reception of control commands from the FCC and forwards the reception status to the PDCU.

Additional sensors are included for data recording and telemetry: A GPS receiver (GNSSU) provides the 3D position for monitoring of the containment of the UAV w.r.t. the boundaries of the operating volume on the ground. An additional inertial measurement unit (BSB) with integrated GPS receiver provides rate and acceleration data and a second GPS position for post flight processing. The omega-dot sensor (ODS) measures angular accelerations. A data concentrator unit (SDCU) forwards these measurements on a common digital data bus. The SDCU also acquires discrete signals that drive the warning LEDs that are normally used to indicate engine warnings in the manned version of the helicopter. The AOX unit acquires the analog signals from a cardan mounted combined angle of attack/angle of sideslip probe that is located on a boom. A central data acquisition unit (DAU) captures and records measurement data from all connected devices and forwards telemetry data to the modem for transmission to the ground station. It is also connected to the helicopter's engine instrumentation: Another data acquisition unit (DAQU) provides measurements of rotor and motor RPM, oil pressure and temperature and other engine parameters. In the manned version of the helicopter, these parameters are provided to the pilot on a display unit (EMSIS) in the cockpit. For the unmanned version, the same parameters are forwarded to the ground, where an identical display unit is mounted in the ground monitoring unit (cf.

Sec. IV.C).

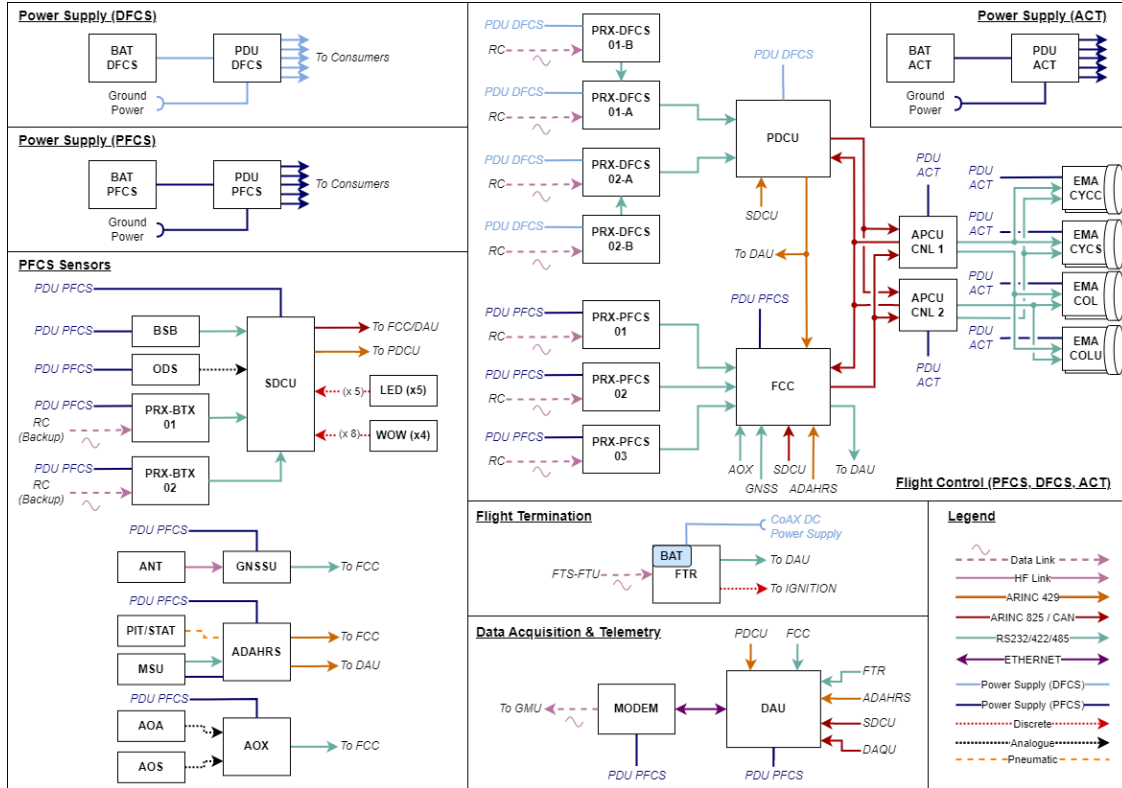


Fig. 16 Onboard flight control system electronic architecture

The containment requirement is addressed by a separate flight termination system, which - completely independent from the FCS - allows to remotely shut down the rotorcraft combustion engine by interrupting the power supply to its ignition circuits. The system uses a separate data link for the transmission of the termination trigger from a ground unit (FTS-FTU) to the airborne receiver (FTR). It is developed to the RCC-319 standard [14], which is based on a robust analogue tone-based modulation of the transmitted command signal. In order to avoid disturbances from interfering RF signals, the datalink uses a restricted frequency band that requires authorization by the German telecommunications authority.

C. Remote Crew Interface

The remote crew interfaces are shown in Fig. 17. Two identical commercial hand-held remote control transmitters are used for the manual control of the rotorcraft by the remote pilot (cf. Fig. 17a). This RC system has been selected due to its redundant data link (each transmitter provides two 2.4 GHz modules and a backup 900 MHz module) and its wide application in the RC model community. In addition to the four main control channels for roll, pitch, yaw and collective control, buttons on the RC are used to select the operating mode of the onboard flight control system. The two centrally located "TAKE OVER" buttons allow to take over control through the backup RC in case of a control malfunction.

A ground monitoring unit (cf. Fig. 17b) provides real-time indication of flight critical system parameters such as the voltage levels of the FCS batteries and the signal strength of the different data links on a main indications panel. This panel also includes an indication of the UA position over a synthetic map view, together with the boundaries of the operating volume. The engine warning LEDs that are installed in the manned version of the helicopter are replicated by software lamps. The panel also includes synoptics sections showing the current operating mode of the flight control system, which are used mainly to provide feedback to the remote pilot during operations and for troubleshooting. Engine parameters and the current fuel level are indicated on a separate display unit (EMSIS), which is identical to the indication unit normally installed in the cockpit of the manned version of the helicopter. Thereby all indications and warnings according to the aircraft flight manual of the manned variant are also provided in the unmanned version.

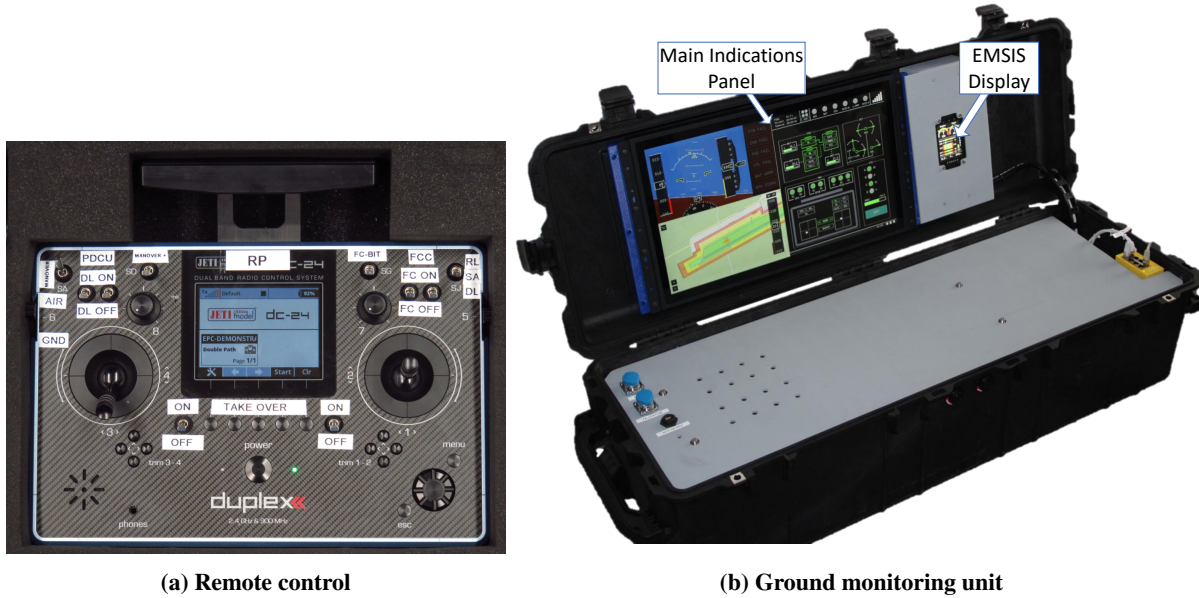


Fig. 17 Remote crew interfaces

D. Validation and Testing

The system architecture design and functions were validated in an integrated system simulation environment that includes the design models for the FCC, PDCU and APCU control software, the GMU display software, sensor models, emulators for the two RCs and the complete digital communication architecture. This integrated system model was developed from a centralized description the logical, transport layer and physical interfaces of the system architecture, stored in a machine readable database format, as described in [15].

Due to the time and budget constraints of the project an exhaustive requirements-based testing of the FCS equipment and the integrated system was not performed. Instead, test cases were derived based on operational sequences that describe the intended system behavior under nominal conditions as well as in failure conditions. This principle is further described in [16]. Those sequences were used to define test cases that test the system behavior in the integrated simulation environment, as well as in hardware-in-the-loop (HIL) tests and ground tests of the integrated system.

Additionally, dedicated ground tests have been performed with the fully integrated system in order to verify the effective range of the data links for remote control, telemetry data transmission and remote flight termination.

V. Flight Tests

The robotic control system was tested in a hardware-in-the-loop test setup first, as already mentioned. After gaining confidence with the system and the flight control functions the robotic actuator unit together with the avionics were mounted into an electric version of the rotorcraft which is installed in a hovering platform. While in the hover frame, the helicopter can take off and land and rotate in the pitch, roll and yaw direction. Limited translational motion was also made possible in the test setup.

The caged flight test campaign was essential to further improve the onboard system and the flight control functions. The opportunity was also used to analyze the effects of the hover frame on the flight dynamics of the helicopter via automatic maneuver injection during the caged flight, in addition to the tests for actuator model identification reported in Sec. III.D. The gathered data during the tests was used for subsequent parameter estimation to identify models of the helicopter flight dynamics during the tests. Fig. 18 shows the hover frame one of our flight tests. The performance of the robotic actuator system was satisfactory in closed loop flight.

In addition to the improvement of the systems and functions, the caged flight tests were an opportunity for the crew to gather experience for the subsequent free flights. The permit-to-fly for free unmanned UAS operation was obtained from the LBA via the SORA process for the SAIL II operation in visual line of sight (VLOS). The flight tests took place at the Magdeburg-Cochstedt airport (EDBC) which is dedicated as a national test center for unmanned aerial vehicles in



Fig. 18 Flight tests in a hover frame.

Germany. The flight volumes approved for operation are visualized in Fig. 19. Three separate areas were cleared for flight, which are the three green overlapping areas of Fig. 19. Due to the size of the airport and our limitation to VLOS flights, a bigger single volume could not be cleared for flight since the aircraft visibility for the pilots might have become insufficient. The vertical extent of the approved flight volume was 50m (140 ft) above ground.



Fig. 19 The UAS operation volumes - each of the overlapping green fields is one volume of operation for VLOS flights.

The flight tests were performed in the attitude mode. One of the two pilots operated the aircraft while a second pilot was on standby and ready to take over in the case of an emergency. The ground crew monitored the flight parameters on a ground station, receiving data from the aircraft via telemetry. The ground station crew maintained communication with the remote pilots via radio intercom. A test director maintained the communication with the airport tower. The helicopter had a flight termination system onboard which was operated by a safety manager. The safety manager's role was to terminate the flight on pilot command, in case of an emergency such as a fly-away outside of the bounds of approved operating volume. Fig. 21 gives an overview of the organization structure of the CoAX UAS crew.

The rotorcraft was flown at hover and slow to middle forward speeds. The system and the flight control law performed as expected in this region of the flight envelope, for which the attitude controller was designed and tested. The helicopter with the closed loop controller could also be operated in windy days with wind velocities up to 15 knots and significant gusts. The pilot could safely operate the helicopter under these conditions without a significant increase in workload. Manual doublets and frequency sweeps were injected during the flight tests to help assess the controller performance and improve it in the future.



Fig. 20 Flight tests in the Magdeburg-Cochstedt airport (EDBC).

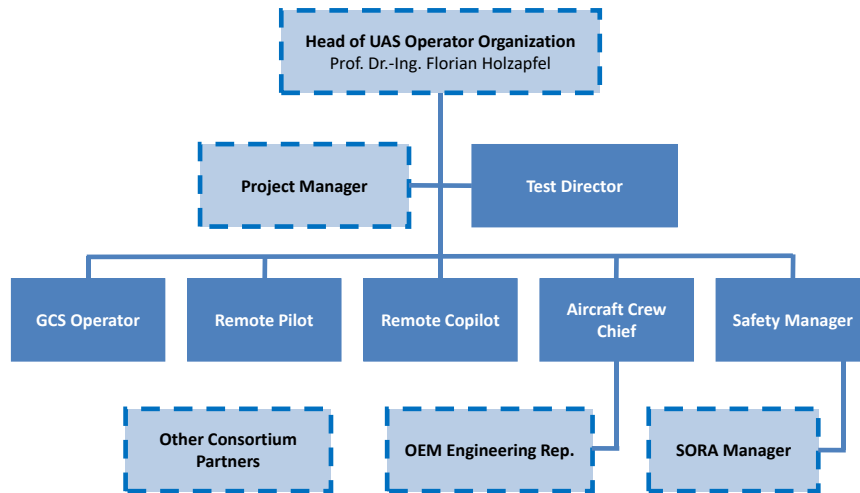


Fig. 21 CoAX 600 UAS crew organization structure.

VI. Conclusion

A 600kg coaxial rotorcraft was converted to a UAV in this project. A robotic actuator system was developed and retrofitted into the cockpit to enable unmanned flight. The robotic actuator system is directly connected to the cockpit inceptors and does not require a modification of the rotorcraft mechanical control system. This minimally invasive approach makes the conversion of a new CoAX 600 rotorcraft to a UAV possible with limited effort and in limited time. The nonlinear kinematics of the robotic system are canceled by the onboard feed-forward robot control software. This is, to the best knowledge of the authors the first time a robotic control system has been used for the unmanned flight of a full-size rotorcraft.

An electronic flight control system, consisting of a primary and an emergency flight control paths was developed and designed and developed, allowing the pilots to control the aircraft using RC transmitters via a primary and an emergency control channel. The onboard system also sent the flight parameters to a ground station via a telemetry link

to monitor the flight tests.

The flight control systems and functions were tested in a HIL setup and later on in a hover frame, before commencing the free flight test campaign. The flight tests took place in the attitude mode flight control law at the national test center for unmanned aircraft in at the Magdeburg-Cochstedt airport (EDBC) in Germany. The remote operation of the rotorcraft was possible with limited pilot workload, despite windy days with significant gusts. The rotorcraft with the closed-loop flight controller showed great flying qualities.

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